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Senior AI Engineer - Agentic Systems, GenAI, RAG, MCP - DevOps, SRE

SUMMARY

Senior Software Engineer with **15+** years building production systems across **full-stack** development, **cloud** infrastructure, and **DevOps/SRE**. Currently focused on **multi-agent AI** platforms, **GenAI**, **RAG** pipelines, and **MCP** infrastructure as a **full-stack Forward Deployed Engineer**. Recent work includes a reusable **MCP** foundation adopted by 8+ engineering teams, a **LangGraph**-based code generation platform, and **AI**-powered **COBOL** modernization. A **Python/Java/TypeScript** and **AWS, Kubernetes & Terraform** foundation enables **end-to-end** ownership of **AI** systems, from **prompt/context** engineering through deployment, evaluation, and **observability**.

EXPERIENCE

Senior Full-Stack FDE & Software Architect · Accenture

Jan 2023 - Present, Columbus, OH

- **MCP Platform foundation:** Built and standardized a reusable **Model Context Protocol** blueprint (**SSE**, **HTTP**, **STDIO** transports) now used by 8+ engineering teams across our AI practice, cutting new **AI** project setup time by ~40% and giving teams a consistent way to expose tools, resources, and prompts to **LLMs**.
- **Multi-Agent Code Generation Platform:** Built a **LangGraph** + **LangChain** + **MCP** system that automates software creation through dynamic, JSON-configurable workflow routing. Clients configure agents, prompts, tasks, and routing without writing code - cutting feature development time by ~30%. Instrumented workflows with **LangSmith** and **Langfuse** for trace-level debugging, prompt regression testing, and quality monitoring in production.
- **COBOL Modernization (RAG + compiler analysis):** Led an **AI** platform that reverse-engineers legacy **COBOL** using **MCP**-orchestrated tools, **AST/CFG/DFG** static analysis, and **RAG** over copybook-resolved source. Generates **Java** implementations and extracts business rules - collapsing weeks of manual analysis into days.
- **BPMN Workflow Documentation Agent:** Built an **LLM** agent that parses XML-based **BPMN/iFlow** workflows into a doubly-linked-list representation, preserving exact execution order, then generates documentation, user stories, and dev artifacts, cutting manual analysis from hours to minutes per integration.. Combined deterministic parsing with **LLM** synthesis to eliminate hallucination on control flow.
- **RAG & Context Engineering:** Designed retrieval pipelines using **OpenAI** embeddings, cascading retrieval (dense + sparse + rerank), and chunking strategies tuned per document type. Established context window management patterns and prompt templates reused across projects.

Stack: *Python • Java • LangChain • LangGraph • LangSmith • MCP • FastMCP • FastAPI • Pinecone • Docker*

Senior Full-Stack Engineer · DevOps & SRE · Accenture

Mar 2019 - Dec 2022, Columbus, OH

- Built a self-healing infrastructure platform (**Java/Spring** + **Angular** + **Grafana/Splunk/Dynatrace**) that cut incident response time by ~45% and minimized manual intervention during outages.
- Authored **Terraform IaC** modules for **AWS (EC2, EKS, VPC, security groups)**, reducing environment provisioning from days to hours.
- Implemented **Prometheus** + **ELK** observability stack with distributed tracing, cutting MTTR by ~40% across services.
- Delivered **SAP BTP/CAP/CPI** event-driven integrations for real-time data ingestion across enterprise systems.

Stack: *Java • Spring Boot • Angular • AWS • Terraform • Kubernetes • Prometheus • ELK • Grafana • SAP BTP*

Senior Full-Stack Engineer · cFive Solutions

Jun 2018 - Mar 2019, Laguna Hills, CA

- Designed a **SAML 2.0** federated authentication platform (**Java** + **Spring Boot** + **Spring Security SAML**) for Los Angeles County, used by the FBI and local law enforcement to access shared county systems.

Stack: *Java • Spring Boot • Spring Security SAML • SAML 2.0 • SSO • Microsoft AAD • MIM • REST • Microservices*

- Built **Hazelcast**-based distributed caching with **Spring AOP** for a high-traffic Ad Network, eliminating concurrency conflicts across servers. Shipped a **Spring Cloud Config** server enabling zero-downtime configuration updates.

Stack: Java • Spring Boot • Spring Cloud Config • Spring AOP • Hazelcast • REST • JUnit • Mockito • Maven

Earlier roles in Brazil (Lojas Americanas, americanas.com.br) building backend systems and integration platforms for retail and e-commerce at scale, 2013–2017

OPEN-SOURCE PROJECTS

MCP Server Language Converter - Hybrid MCP server for legacy code analysis - [GitHub - MCP](#)

- Open-source MCP server exposing 11+ COBOL reverse-engineering tools (AST, ASG, CFG, DFG, complexity, copybook resolution, call-graph, system-wide dependency analysis) for LLM-driven legacy modernization. Dual-interface: MCP protocol (STDIO, SSE, Streamable HTTP) via FastMCP 2.0;
- Hexagonal (ports & adapters) architecture with JSON-config-driven tool registration enabling zero code duplication when adding new domain servers. Integrates with LangGraph for multi-agent COBOL modernization workflows. Python 3.12, PostgreSQL, async SQLAlchemy, UV.

MCP Infra, AWS - Terragrunt IaC deploying the MCP Server to AWS - [GitHub - Infra](#)

- Production-grade Terragrunt IaC deploying the MCP Server to AWS (EC2, Elastic IP, IAM, Route 53) with S3-backed remote state and DynamoDB locking. Hierarchical config (account → environment → region → service) enables clean multi-environment expansion. Makefile lifecycle automation; EC2 bootstrap runs Docker Compose (PostgreSQL, MCP server, NGINX) with optional Let's Encrypt SSL.

DocWise - RAG-powered documentation assistant - [GitHub - DocWise](#)

- Production RAG application that ingests documentation from any URL and serves context-aware conversational answers. Built on a protocol-based, swappable architecture (models, embedding strategies, chunkers, and vector stores configurable without code changes), with LangChain, GPT-4o-mini, OpenAI embeddings, dual vector store (Chroma for dev, Pinecone for prod, 1536d cosine), and a TavilyExtract-first ingestion pipeline with automatic BeautifulSoup fallback; Implements cascading retrieval (dense + sparse + rerank), conversational memory, and full LangSmith tracing for prompt debugging and quality monitoring. Streamlit UI.

iFlow XML Parser (SAP CAP + GenAI) - [GitHub - iFlow GenAI](#)

- SAP CAP application (TypeScript + Node.js + CDS, deployed to SAP BTP via MTA) that parses SAP Integration Suite iFlow XML into a doubly-linked-list step graph, preserving exact execution order, then uses generative AI (OpenAI) to produce structured documentation. Exposes both REST and SOAP interfaces (WSDL-driven).

TECHNICAL SKILLS

AI/ML & Retrieval - LangChain, LangGraph, LangSmith, Langfuse, MCP, RAG, multi-agent systems, agentic workflows, prompt & context engineering, tool / function calling, LLM evaluation, OpenAI, AWS Bedrock, Pinecone, Chroma, embeddings, cascading retrieval, Tavily, Claude, Copilot;

Languages - Python, Java, TypeScript, JavaScript; **Backend** - FastAPI, Flask, Spring Boot, REST, SOAP, microservices, event-driven architecture; **Frontend** - React, Angular, Streamlit, TypeScript;

Cloud & Infrastructure - AWS (Bedrock, EC2, EKS, Lambda), Azure, GCP, Kubernetes, Docker, Terraform, Helm;

DevOps & Observability - Git, CI/CD, Grafana, Splunk, Prometheus, Dynatrace, ELK.

CERTIFICATIONS

Full Stack FDE - Forward Deployed Engineer, Accenture (Credly-verified) • CKAD - Certified Kubernetes Application Developer • HashiCorp Terraform Associate • AWS Certified SysOps Administrator • AWS Certified Developer Associate • CFCD - Cloud Foundry Certified Developer

EDUCATION

Professional Specialization in Software Engineering - Universidade Federal de Minas Gerais, Brazil

Bachelor of Technology in Data Processing - Centro Universitário UNA, Brazil